Electricity and magnetism

120 questions: 60 multiple choice, 60 structured.

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1 · Part 1

Core | Simple phenomena of magnetism | [1]

A toy train has carriages that are joined by small bar magnets fixed to the ends of each carriage. The magnet at the back of carriage X has its N pole facing backwards. Carriage Y is pushed towards the back of carriage X, so that the magnet at the front of Y comes close to the magnet at the back of X.

Which pole of the magnet at the front of Y must face carriage X so that the two carriages pull together?

A the N pole

B the S pole

C either pole, because all magnets attract each other

D neither pole, because magnets only attract unmagnetised materials

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2 · Part 1

Core | Simple phenomena of magnetism | [1]

The two ends of a metal bar are brought, in turn, close to each pole of a freely suspended bar magnet.

Which observation shows that the metal bar is itself a magnet?

A One end of the bar attracts the N pole of the suspended magnet.

B Both ends of the bar attract both poles of the suspended magnet.

C Neither end of the bar has any effect on the suspended magnet.

D One end of the bar repels the N pole of the suspended magnet.

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3 · Part 1

Core | Simple phenomena of magnetism | [1]

Which statement about soft iron and steel is correct?

A Soft iron is used for temporary magnets because it loses its magnetism easily.

B Steel is used for temporary magnets because it loses its magnetism easily.

C Soft iron is used for permanent magnets because it keeps its magnetism.

D Steel loses all its magnetism as soon as it is taken away from a strong magnet.

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4 · Part 1 Core Simple phenomena of magnetism [1]

Which list contains only magnetic materials?

A copper, iron, nickel

B aluminium, cobalt, steel

C cobalt, iron, nickel

D brass, iron, steel

5 · Part 1 Core Simple phenomena of magnetism [1]

A bar magnet lies on a bench pointing east–west, with its N pole at the east end. A small plotting compass is placed on the bench a short distance to the east of the N pole, on the line through the length of the magnet. Ignore any other magnetic fields.

In which direction does the N pole of the compass needle point?

A east

B west

C north

D south

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6 · Part 1 Core Simple phenomena of magnetism [1]

In which device must the magnet be an electromagnet rather than a permanent magnet?

A the seal that keeps a refrigerator door closed

B the needle of a plotting compass

C a crane that lifts scrap steel and then drops it into a lorry

D the catch that holds a cupboard door shut

7 · Part 1 Extended Simple phenomena of magnetism [1]

A student plots the magnetic field lines around a bar magnet. Near point P the field lines are about 2 mm apart. Near point Q the field lines are about 6 mm apart.

Which statement is correct?

A The magnetic field is stronger at Q than at P.

B The magnetic field is stronger at P than at Q.

C The magnetic field is equally strong at P and at Q, because both points are near the same magnet.

D The spacing of the field lines gives no information about the strength of the field.

8 · Part 1 Core Simple phenomena of magnetism [1]

A bar magnet is held vertically with its N pole at the bottom. An unmagnetised iron nail hangs from the N pole, with the head of the nail touching the pole. A second unmagnetised iron nail hangs from the point of the first nail.

Which statement explains why the second nail is held up?

A The first nail becomes a permanent magnet as soon as it touches the bar magnet.

B The bar magnet attracts the second nail directly, and the first nail plays no part.

C Both nails become electrically charged when they touch each other.

D Magnetism is induced in the first nail, with an S pole at its head and an N pole at its point.

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9 · Part 1 Core Electric charge [1]

Three small spheres X, Y and Z hang from insulating threads. Every sphere is electrically charged. When X is brought near Y, they repel each other. When Y is brought near Z, they attract each other. Sphere Z is positively charged.

What are the charges on X and on Y?

	charge on X	**charge on Y**
A	negative	negative
B	positive	positive
C	positive	negative
D	negative	positive

10 · Part 1 Core Electric charge [1]

A polythene rod is rubbed with a dry cloth. The rod becomes negatively charged.

Which statement describes what happens during the rubbing?

A Positive charges move from the rod to the cloth.

B Positive charges move from the cloth to the rod.

C Electrons move from the rod to the cloth.

D Electrons move from the cloth to the rod.

11 · Part 1 Core Electric charge [1]

Which statement explains why copper is a good electrical conductor?

A All the electrons in copper are held tightly to their atoms.

B Copper contains free protons that can move through the metal.

C Copper contains free electrons that can move through the metal.

D The atoms of copper have no electrons.

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12 · Part 1 Extended Electric charge [1]

Which statement describes an electric field?

A. a region in which a magnetic pole experiences a force

B. a region in which an electric charge experiences a force

C. the flow of electric charge around a circuit

D. a region in which any mass experiences a force

13 · Part 1 Extended Electric charge [1]

How is the direction of an electric field at a point defined?

A. the direction of the force on a negative charge at that point

B. the direction of the force on a positive charge at that point

C. the direction in which free electrons move at that point

D. the direction of the force on the N pole of a magnet at that point

14 · Part 1 Extended Electric charge [1]

A metal sphere stands on an insulating support, well away from other objects. The sphere is given a negative charge.

Which statement describes the electric field lines outside the sphere?

A. parallel, evenly spaced lines all pointing in the same direction

B. straight radial lines pointing away from the sphere

C. circles around the sphere

D. straight radial lines pointing towards the sphere

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15 · Part 1 Extended Electric charge [1]

A small positive point charge is fixed in place. A tiny negatively charged particle is held at point Y, a short distance from the point charge.

What is the direction of the electric field at Y, and what is the direction of the electric force on the particle?

	direction of electric field at Y	direction of force on particle
A	away from the point charge	away from the point charge
B	towards the point charge	towards the point charge
C	away from the point charge	towards the point charge
D	towards the point charge	away from the point charge

16 · Part 1 Core Simple phenomena of magnetism [4]

A student has three rods that look the same size and shape. One is made of aluminium, one is unmagnetised iron and one is a steel bar magnet. The rods are not labelled.

(a) State the difference between a magnetic material and a non-magnetic material. [1]

(b) The student holds the N pole of a second bar magnet close to each end of each rod in turn.

Describe what the student observes for each of the three rods. [3]

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17 · Part 1 Core Simple phenomena of magnetism [6]

A technician tests two unmagnetised rods of the same size. Rod X is made of soft iron and rod Y is made of steel. Each rod is held vertically with its top end touching the S pole of a strong magnet. The bottom end of each rod is then dipped into a tray of small iron tacks. Both rods pick up tacks.

(a) Explain why the rods pick up tacks, even though they were unmagnetised at the start. [2]

(b) State which magnetic pole is formed at the top end of each rod, where it touches the S pole of the magnet. [1]

(c) The strong magnet is then taken away from both rods.

Describe and explain what happens to the tacks hanging from rod X and from rod Y. [3]

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18 · Part 1 Core Simple phenomena of magnetism [6]

A student wants to find the shape of the magnetic field around a bar magnet. She has a bar magnet, a sheet of paper, a pencil and a small plotting compass.

(a) Describe what is meant by a magnetic field. [1]

(b) Describe how the student uses the plotting compass to plot one magnetic field line, including its direction, starting near the N pole of the magnet. [4]

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(c) Another student uses iron filings sprinkled on the paper instead of a plotting compass.

State one piece of information about the field that the iron filings do not show. [1]

19 · Part 1 Core Simple phenomena of magnetism [6]

At a recycling centre, a crane with a large electromagnet lifts steel cans out of a heap of mixed steel and aluminium cans, and drops them into a separate bin.

(a) Explain why the aluminium cans are left behind in the heap. [1]

(b) State one advantage of using an electromagnet, rather than a permanent magnet, on this crane. [1]

(c) State two uses of permanent magnets. [2]

(d) The core of the electromagnet is made of soft iron, not steel.

Suggest why soft iron is used for the core. [2]

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20 · Part 1 Extended Simple phenomena of magnetism [5]

Two bar magnets, P and Q, lie in a straight line on a bench. The N pole of P is 4 cm from the S pole of Q. The magnets attract each other.

(a) Explain, in terms of magnetic fields, why the magnets exert forces on each other. [2]

(b) A student plots the field lines. Between the two magnets the field lines are close together. Beyond the far ends of the magnets the field lines are much further apart.

State what this shows about the magnetic field. [1]

(c) Magnet Q is moved so that the gap between the N pole of P and the S pole of Q is 8 cm.

Predict how the spacing of the field lines in the gap and the force between the magnets change. Give a reason for your answer. [2]

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21 · Part 1 Core Simple phenomena of magnetism [6]

A desk toy has two ring magnets threaded onto a vertical wooden rod. The lower ring rests on the base of the toy. The upper ring floats at rest above the lower ring without touching it. Each ring has one pole on its top face and the other pole on its bottom face. The top face of the lower ring is an N pole.

(a) State which pole is on the bottom face of the upper ring. [1]

(b) Explain why the upper ring stays at rest above the lower ring. [2]

(c) A child pushes the upper ring down a little way and then lets go.

Suggest what happens to the upper ring, and why. [2]

(d) The lower ring is taken off the rod, turned upside down and put back. The upper ring is put back on top.

Describe what happens to the upper ring. [1]

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22 · Part 1 Core Simple phenomena of magnetism [4]

A magnetic catch keeps a cupboard door closed. A permanent magnet is fixed to the cupboard frame, and a small metal plate is fixed to the door. The plate is not a magnet.

(a) Suggest a suitable metal for the plate. [1]

(b) Explain why the plate is attracted to the magnet, even though the plate is not a magnet. [2]

(c) State why the magnet on the frame is made of steel rather than soft iron. [1]

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23 · Part 1 Core Electric charge [5]

A child rubs a rubber balloon on a woollen jumper. The balloon becomes negatively charged.

(a) State the two types of electric charge. [1]

(b) Explain how the balloon becomes negatively charged. Your answer should name the particles that move. [2]

(c) State the charge on the woollen jumper after the rubbing, and explain your answer. [2]

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24 · Part 1 Core Electric charge [5]

A student has two identical plastic rulers, a dry woollen cloth, a length of thread and some small pieces of tissue paper.

(a) Describe how the student can charge one ruler by friction and then show that it is charged, using the tissue paper. [3]

(b) The student charges both rulers in the same way.

Describe how the student can use the thread to show that the two rulers carry the same type of charge. [2]

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25 · Part 1 Core Electric charge [7]

A student wants to sort four samples into electrical conductors and electrical insulators. The samples are a steel nail, a plastic ruler, a piece of copper wire and a glass rod. She has a 3.0 V battery, a small lamp, three connecting leads and two crocodile clips.

(a) Describe how the student connects the apparatus and uses it to test each sample. [3]

(b) State which of the four samples are conductors and which are insulators. [2]

(c) Use a simple electron model to explain why the steel nail conducts electricity but the glass rod does not. [2]

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26 · Part 1 Core Electric charge [5]

A student holds a plastic rod in her hand and rubs it with a dry cloth. The plastic rod becomes charged. She then holds a copper rod in her hand and rubs it with the same cloth in the same way. The copper rod does not stay charged.

(a) Explain, using a simple electron model, why the charge stays on the plastic rod. [2]

(b) Explain why the copper rod does not stay charged while the student holds it in her hand. [2]

(c) Suggest one change the student could make so that the copper rod stays charged after rubbing. [1]

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27 · Part 1 Extended Electric charge [5]

(a) State the unit in which electric charge is measured. [1]

(b) At point P, the electric field is directed due east.

(i) State the direction of the electric force on a small positive charge placed at P. [1]

(ii) State the direction of the electric force on an electron placed at P. [1]

(c) Compare the way the direction of an electric field at a point is defined with the way the direction of a magnetic field at a point is defined. [2]

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28 · Part 1 Extended Electric charge [6]

Two flat metal plates are fixed horizontally, one directly above the other, with a gap between them. The upper plate is connected to the positive terminal of a high-voltage supply and the lower plate is connected to the negative terminal.

(a) Describe the pattern and the direction of the electric field in the space between the plates, away from their edges. [2]

(b) A tiny oil drop with a negative charge is in the space between the plates.

State the direction of the electric force on the drop. Explain your answer. [2]

(c) The drop is moved to a different position between the plates, closer to the lower plate but still well away from the edges.

State whether the electric force on the drop changes. Give a reason for your answer. [2]

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29 · Part 1 Extended Electric charge [4]

In a science museum, a large metal sphere stands on a tall insulating column. The sphere is given a positive charge. There are no other charged objects nearby.

(a) Describe the pattern and the direction of the electric field lines outside the sphere. [2]

(b) A tiny dust particle with a negative charge floats in the air near the sphere.

State the direction of the electric force on the dust particle. Explain your answer in terms of the direction of the electric field. [2]

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30 · Part 1 Core Electric charge [7]

A girl combs her clean, dry hair with a plastic comb. Afterwards the comb is negatively charged and the strands of her hair stand apart from each other. Before combing, neither the comb nor the hair had any overall charge.

(a) State the type of charge on the hair after combing. [1]

(b) Explain, in terms of the movement of charged particles, how the comb and the hair become charged. [2]

(c) Explain why the strands of hair stand apart from each other. [2]

(d) State the total charge of the comb and the hair taken together after combing. Explain your answer. [2]

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1 · Part 2 Core Electric current [1]

There is a current in a copper wire that is connected to a battery.

Which statement describes how the copper wire conducts electricity?

A Positive ions move through the copper.

B Free electrons move through the copper.

C Protons move through the copper.

D The copper atoms move along the wire.

2 · Part 2 Core Electric current [1]

Which statement describes an alternating current (a.c.)?

A It always flows in the same direction and has a steady value.

B It flows in one direction only, but its size rises and falls.

C It is the type of current produced by a single cell.

D It repeatedly reverses its direction.

3 · Part 2 Core Electric current [1]

The current in a circuit is expected to be about 0.3 A. A student's ammeter has four ranges.

Which range should she use to get the most precise reading without the current going beyond the end of the scale?

A 0 to 1 A

B 0 to 10 A

C 0 to 100 mA

D 0 to 10 mA

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4 · Part 2 Extended Electric current [1]

In a time of 2.0 minutes, a charge of 36 C passes a point in a wire.

What is the current in the wire?

A 18 A

B 3.3 A

C 0.30 A

D 4300 A

5 · Part 2 Core Electromotive force and potential difference [1]

Which statement defines the electromotive force (e.m.f.) of a source?

A the electrical work done by the source in moving a unit charge around a complete circuit

B the work done by a unit charge passing through a component

C the amount of charge that the source moves around the circuit each second

D the energy that the source transfers each second

6 · Part 2 Core Electromotive force and potential difference [1]

In a circuit, a battery, a switch, an ammeter and a lamp are connected in series. A student wants to measure the potential difference (p.d.) across the lamp.

How should she connect the voltmeter?

A in series with the lamp

B in parallel with the lamp

C in parallel with the ammeter

D in series with the switch

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7 · Part 2 Extended Electromotive force and potential difference [1]

As a charge of 15 C passes through an electric heater, the charge does 3450 J of work.

What is the potential difference (p.d.) across the heater?

A 0.0043 V

B 23 V

C 52 000 V

D 230 V

8 · Part 2 Core Electromotive force and potential difference [1]

Which row gives the unit of electromotive force (e.m.f.) and the unit of potential difference (p.d.)?

	unit of e.m.f.	**unit of p.d.**
A	ampere	volt
B	volt	ampere
C	volt	volt
D	joule	volt

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9 · Part 2 Extended Electromotive force and potential difference [1]

In a circuit, a battery of e.m.f. 9.0 V drives a current of 0.20 A for 50 s.

How much electrical work does the battery do in this time?

A 10 J

B 90 J

C 450 J

D 2250 J

10 · Part 2 Core Resistance [1]

The potential difference (p.d.) across a resistor is 6.0 V. The current in the resistor is 0.40 A.

What is the resistance of the resistor?

A 15 Ω

B 2.4 Ω

C 0.067 Ω

D 6.4 Ω

11 · Part 2 Core Resistance [1]

Four wires are made of the same metal and are at the same temperature.

Which wire has the greatest resistance?

	length of wire	**thickness of wire**
A	short	thick
B	short	thin
C	long	thick
D	long	thin

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12 · Part 2 Extended Resistance [1]

The resistance of a metal wire is 12 Ω. A second wire is made of the same metal. It has half the length and three times the cross-sectional area of the first wire.

What is the resistance of the second wire?

A 72 Ω

B 18 Ω

C 2.0 Ω

D 8.0 Ω

13 · Part 2 Core Electrical energy and electrical power [1]

The lamp in a bicycle light is connected to a 4.5 V battery. The current in the lamp is 0.60 A.

What is the power of the lamp?

A 2.7 W

B 7.5 W

C 0.13 W

D 5.1 W

14 · Part 2 Core Electrical energy and electrical power [1]

Which statement describes one kilowatt-hour (kW h)?

A the power used by a 1 kW appliance in 1 hour

B the energy transferred by a 1 kW appliance in 1 hour

C the energy transferred by a 1 kW appliance in 1 second

D a rate of energy transfer of 1000 J every hour

15 · Part 2 Core Electrical energy and electrical power [1]

An electric heater of power 2.0 kW is switched on for 90 minutes every day for 7 days. Electricity costs 30 cents per kW h.

What is the total cost of using the heater for the 7 days?

A 90 cents

B 378 cents

C 630 cents

D 37 800 cents

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16 · Part 2

Core

Electric current

[4]

In a circuit, a length of copper wire is connected across the terminals of a cell, and there is a current in the wire.

(a) State what an electric current is a flow of. [1]

(b) Describe how the copper wire conducts electricity. Your answer should refer to the particles in the metal. [2]

(c) The cell is replaced by a power supply that provides an alternating current (a.c.). State how the movement of the free electrons in the wire is now different from when the cell was connected. [1]

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17 · Part 2 Core Electric current [4]

In an experiment, a student wants to measure the current in a small electric motor. The motor is connected to a battery and a switch.

(a) State how the ammeter must be connected to the motor. [1]

(b) The student has an analogue ammeter with two ranges: 0 to 1.0 A and 0 to 5.0 A. She expects the current in the motor to be between 0.40 A and 0.80 A.

State which range she should use. Explain your choice. [2]

(c) The student's teacher gives her a digital ammeter instead. Suggest one advantage of using a digital ammeter rather than an analogue ammeter. [1]

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18 · Part 2 Extended Electric current [6]

While a phone is charging, its charger is connected to it by a cable, and there is a steady current in the cable.

(a) Define electric current. [1]

(b) The current in the cable is 1.5 A. The phone charges for 40 minutes.

Calculate the charge that flows through the cable in this time. [3]

(c) The charger has a positive terminal and a negative terminal.

(i) State the direction of the conventional current in the external circuit, in terms of these terminals. [1]

(ii) State the direction in which the free electrons flow in the external circuit, in terms of these terminals. [1]

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19 · Part 2 Extended Electric current [6]

When a camera flash fires, a charge of 0.024 C flows through the flash tube in a time of 1.2 ms. The charge is carried by electrons.

(a) Calculate the average current in the flash tube. [3]

(b) The charge on one electron is 1.6×10^{-19} C.

Calculate the number of electrons that flow through the flash tube during the flash. [2]

(c) In a different flash setting, the same charge flows through the tube in 4.8 ms.

Deduce the average current in the tube for this setting. [1]

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20 · Part 2

Core

Electromotive force and potential difference

[5]

Inside a torch, a battery is connected in a circuit with a switch and a lamp.

(a) Define the potential difference (p.d.) across the lamp. [2]

(b) State the unit of p.d. [1]

(c) The battery is labelled "3.0 V". This value is the electromotive force (e.m.f.) of the battery.

Explain what this value tells you about the battery, in terms of work and charge. [2]

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21 · Part 2 Core Electromotive force and potential difference [4]

In an experiment, a student uses an analogue voltmeter to measure the potential difference (p.d.) across a resistor. The voltmeter has two ranges, 0 to 3 V and 0 to 15 V, which share the same pointer.

(a) State how the voltmeter is connected to the resistor. [1]

(b) The student first sets the voltmeter to the 0 to 15 V range. The reading is 2.4 V.

State which range she should now use. Explain why. [2]

(c) Later, the p.d. across the resistor is increased to about 9 V.

State what would happen if the student kept the voltmeter on the 0 to 3 V range. [1]

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22 · Part 2 Extended Electromotive force and potential difference [6]

The p.d. across a car headlamp is 12 V. In a time of 5.0 minutes, a charge of 1500 C passes through the headlamp.

(a) Calculate the energy transferred by the charge as it passes through the headlamp in this time. [2]

(b) Calculate the current in the headlamp. [3]

(c) State the energy transferred in the headlamp by each coulomb of charge that passes through it. [1]

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23 · Part 2 Extended Electromotive force and potential difference [8]

In a circuit, a battery of e.m.f. 6.0 V is connected to a resistor by two long connecting leads. In a time of 80 s, the battery does 120 J of electrical work moving charge around the complete circuit. During this time, the p.d. across the resistor is 5.5 V.

(a) Calculate the charge that the battery moves around the circuit in 80 s. [2]

(b) Calculate the current in the circuit. [2]

(c) Calculate the energy transferred in the resistor in 80 s. [2]

(d) Calculate the energy transferred in the rest of the circuit in 80 s, and suggest where in the circuit this energy is transferred. [2]

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24 · Part 2 Core Electromotive force and potential difference [4]

In a simple circuit, a solar cell is connected to a small electric motor. The motor turns.

(a) State one similarity between the electromotive force (e.m.f.) of the solar cell and the potential difference (p.d.) across the motor. [1]

(b) State one difference between e.m.f. and p.d. [1]

(c) A student says: "The current gets used up in the motor."

Explain why the student is wrong. [2]

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25 · Part 2 Core Resistance [7]

In an experiment, a student is asked to determine the resistance of a length of metal wire. She has a battery, a switch, an ammeter, a voltmeter, a variable resistor and connecting leads.

(a) Describe how she should connect the circuit and what readings she should take. [4]

(b) The table shows her results.

p.d. / V	current / A
1.0	0.20
2.0	0.41
3.0	0.59
4.0	0.80

Use the reading at 4.0 V to calculate the resistance of the wire. [2]

(c) Suggest one reason why she takes several sets of readings instead of only one. [1]

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26 · Part 2 Core Resistance [5]

Inside an electric toy, a thin metal wire is connected to a 9.0 V battery. The current in the wire is 0.36 A.

(a) Calculate the resistance of the wire. [2]

(b) The manufacturer replaces the wire with a thicker wire. The new wire is made of the same metal and has the same length.

(i) State how the resistance of the thicker wire compares with the resistance of the original wire. [1]

(ii) The thicker wire is connected to the same 9.0 V battery. State how the current in the thicker wire compares with 0.36 A. [1]

(c) State how the resistance changes if a longer wire of the same metal and the same thickness is used. [1]

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27 · Part 2 Extended Resistance [7]

The resistance of a 2.0 m length of nichrome wire is 8.0 Ω.

(a) Calculate the resistance of a 1.5 m length of the same wire. [2]

(b) A second nichrome wire is 2.0 m long, but its diameter is half the diameter of the first wire.

Calculate the resistance of the second wire. [3]

(c) The 1.5 m length of wire from (a) is connected across a p.d. of 3.0 V.

Calculate the current in the wire. [2]

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28 · Part 2

Core

Electrical energy and electrical power

[6]

An electric fan is connected to a 24 V supply. The current in the fan motor is 0.75 A.

(a) Describe how energy is transferred from the supply to the surroundings while the fan is running. [2]

(b) Calculate the power of the fan. [2]

(c) The fan runs for 25 minutes. Calculate the electrical energy transferred by the fan in this time. [2]

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29 · Part 2 Core Electrical energy and electrical power [7]

The table shows three electrical appliances in a home, their power and the time each one is used every day.

appliance	power / W	time used each day / h
oven	3000	0.50
television	120	5.0
lights (all together)	400	6.0

(a) Define the kilowatt-hour (kW h). [1]

(b) Calculate the total energy used by the three appliances each day, in kW h. [2]

(c) Electricity costs 28 cents per kW h. Calculate the cost, in cents, of using the three appliances for 30 days. [2]

(d) Calculate the energy used by the oven each day, in joules. [2]

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30 · Part 2 Core Electrical energy and electrical power [8]

An electric kettle is designed for a 240 V mains supply. Its power is 2.4 kW.

(a) Calculate the current in the kettle when it is connected to the 240 V supply. [2]

(b) Calculate the resistance of the heating element. [2]

(c) The kettle is switched on for 2.5 minutes. Calculate the electrical energy transferred. [2]

(d) The kettle is taken to a country where the mains supply is 120 V. Assume the resistance of the heating element does not change.

Calculate the power of the kettle on the 120 V supply. [2]

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1 · Part 3 Core Series and parallel circuits [1]

In a circuit, a battery and three lamps are connected in series. An ammeter connected between the battery and the first lamp reads 0.30 A. A second ammeter is connected between the second lamp and the third lamp.

What does the second ammeter read?

A 0.10 A

B 0.30 A

C 0.15 A

D 0.90 A

2 · Part 3 Core Series and parallel circuits [1]

Four cells are connected in series by a student. Each cell has an e.m.f. of 1.5 V. By mistake, one of the four cells is connected the wrong way round.

What is the combined e.m.f. of the four cells?

A 1.5 V

B 4.5 V

C 3.0 V

D 6.0 V

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3 · Part 3 Core Series and parallel circuits [1]

Three resistors of resistance 12 Ω, 18 Ω and 30 Ω are connected in series.

What is their combined resistance?

A 60 Ω

B 30 Ω

C 20 Ω

D 5.8 Ω

4 · Part 3 Core Series and parallel circuits [1]

Two resistors, of resistance 10 Ω and 40 Ω, are connected in parallel with each other.

Which value could be the combined resistance of the two resistors?

A 8.0 Ω

B 25 Ω

C 50 Ω

D 400 Ω

5 · Part 3 Core Series and parallel circuits [1]

The lamps in a house lighting circuit are connected in parallel with each other.

What is an advantage of connecting the lamps in parallel?

A The combined resistance of the lamps is larger than the resistance of one lamp.

B The current from the supply is smaller than the current in each lamp.

C Each lamp has a smaller p.d. across it, so it lasts longer.

D Each lamp can be switched on and off without affecting the other lamps.

6 · Part 3 Extended Series and parallel circuits [1]

At a junction in a circuit, two wires carry currents of 0.50 A and 0.20 A into the junction. Three wires carry current away from the junction. Two of these wires carry currents of 0.15 A and 0.25 A.

What is the current in the third wire leaving the junction?

A 0.10 A

B 0.30 A

C 0.40 A

D 1.10 A

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7 · Part 3

Extended | Series and parallel circuits | [1]

Two resistors, of resistance 6.0 Ω and 3.0 Ω, are connected in parallel with each other across a 12 V power supply. Ignore the resistance of the supply and the connecting wires.

What is the current from the supply?

A 1.3 A

B 2.0 A

C 6.0 A

D 4.0 A

8 · Part 3

Core | Action and use of circuit components | [1]

Three resistors, of resistance 5.0 Ω, 10 Ω and 20 Ω, are connected in series with a power supply. There is a constant current in the circuit.

Across which resistor is the p.d. largest?

A the 5.0 Ω resistor

B the 10 Ω resistor

C the p.d. is the same across all three resistors

D the 20 Ω resistor

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9 · Part 3

Extended | Action and use of circuit components | [1]

Two resistors, R1 and R2, are connected in series across a 9.0 V supply to form a potential divider. R1 has a resistance of 2.0 kΩ. The p.d. across R2 is 6.0 V.

What is the resistance of R2?

A 4.0 kΩ

B 1.0 kΩ

C 3.0 kΩ

D 6.0 kΩ

10 · Part 3 Core Electrical safety [1]

An electric iron is connected to the 230 V mains supply. The power of the iron is 1610 W. Fuses rated at 3 A, 5 A, 10 A and 13 A are available.

Which fuse is the most suitable for the plug of the iron?

A 3 A

B 5 A

C 10 A

D 13 A

11 · Part 3 Core Electrical safety [1]

Why must the switch for a mains appliance be connected in the live wire?

A because the live wire carries a larger current than the neutral wire

B so that, when the switch is open, the appliance is no longer connected to the high p.d. of the live wire

C so that the fuse melts each time the switch is opened

D so that the earth wire carries the current when the appliance is switched off

12 · Part 3 Core Electrical safety [1]

Why is it hazardous to use a mains electrical appliance in damp conditions?

A Water causes the insulation on the cable to melt.

B Water makes the fuse melt, so the appliance stops working.

C Water increases the resistance of the cable, so the cable overheats.

D Water can conduct a current through a person, causing an electric shock.

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13 · Part 3 Core Electromagnetic induction [1]

The ends of a coil of wire are connected to a sensitive centre-zero ammeter. A bar magnet is used near the coil.

In which situation is there **no** induced e.m.f. in the coil?

A The magnet is pushed into the coil.

B The coil is moved towards the magnet, which is held still.

C The magnet is held at rest inside the coil.

D The magnet is pulled out of the coil.

14 · Part 3 Core Electromagnetic induction [1]

In an experiment, a bar magnet is pushed into a coil and an e.m.f. is induced in the coil.

Which change increases the magnitude of the induced e.m.f.?

A using a coil with fewer turns

B using a stronger magnet

C pushing the magnet in more slowly

D turning the magnet round so that the other pole enters the coil first

15 · Part 3 Extended Electromagnetic induction [1]

In a uniform magnetic field that is directed vertically downwards, a straight horizontal wire lies along a north–south line. The ends of the wire are connected to a meter by long flexible leads that stay outside the field.

The wire is moved horizontally towards the east, staying along the north–south line.

In which direction is the induced current in the wire?

A towards the north

B towards the south

C vertically upwards

D towards the west

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16 · Part 3 Core Series and parallel circuits [5]

Three cells, each of e.m.f. 1.5 V, are connected in series, all pointing the same way. They are connected in series with a 2.0 Ω resistor and a 7.0 Ω resistor. Ignore the resistance of the cells and the connecting wires.

(a) Calculate the combined e.m.f. of the three cells. [1]

(b) Calculate the combined resistance of the two resistors. [1]

(c) Calculate the current in the circuit. [2]

(d) State the current in the 2.0 Ω resistor. Give a reason for your answer. [1]

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17 · Part 3 Core Series and parallel circuits [6]

A house lighting circuit has four identical lamps connected in parallel with each other to the mains supply. Each lamp has its own switch.

(a) State two advantages of connecting the lamps in parallel rather than in series. [2]

(b) When all four lamps are switched on, the current in each lamp is 0.25 A. State whether the current from the supply is larger than, equal to or smaller than 0.25 A. [1]

(c) Only two of the lamps are switched on. State how the combined resistance of these two lamps compares with the resistance of one lamp. [1]

(d) Explain why the current from the supply increases when more of the lamps are switched on. [2]

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18 · Part 3 Core Series and parallel circuits [5]

A student builds a series circuit containing a battery, a switch, a lamp and a resistor.

(a) Describe how the student should connect an ammeter to measure the current in the lamp, and a voltmeter to measure the p.d. across the resistor. [2]

(b) The student closes the switch and connects the ammeter at three different positions in the circuit. The table shows the results.

position of ammeter	current / A
between the battery and the switch	0.24
between the switch and the lamp	0.24
between the lamp and the resistor	0.24

State what these results show about the current in a series circuit. [1]

(c) The student adds a second resistor in series with the first resistor. The battery is unchanged.

(i) State what happens to the combined resistance of the circuit. [1]

(ii) State what happens to the current in the lamp. [1]

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19 · Part 3 Extended Series and parallel circuits [7]

A 10 V power supply is connected in series with an 8.0 Ω resistor and a parallel arrangement. The parallel arrangement is a 20 Ω resistor and a 30 Ω resistor connected in parallel with each other. Ignore the resistance of the supply and the connecting wires.

(a) Calculate the combined resistance of the 20 Ω and 30 Ω resistors in parallel. [2]

(b) Calculate the total resistance of the circuit. [1]

(c) Calculate the current from the power supply. [1]

(d) Calculate the p.d. across the 8.0 Ω resistor and the p.d. across the parallel arrangement. [2]

(e) Calculate the current in the 30 Ω resistor. [1]

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20 · Part 3 Extended Series and parallel circuits [4]

A current of 2.4 A flows into a junction J. At J the current divides between three resistors, X, Y and Z, which are connected in parallel with each other. The current in X is 0.80 A and the current in Y is 0.60 A.

(a) Explain why the total current into junction J is equal to the total current out of J. [2]

(b) Calculate the current in Z. [1]

(c) The p.d. across each of the three resistors is the same. Deduce which resistor has the largest resistance. Give a reason for your answer. [1]

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21 · Part 3 Extended Series and parallel circuits [6]

A car seat heater has two heating elements. Each element has a resistance of 24 Ω. The heater is connected to a 12 V car battery. A switch connects the two elements either in series with each other (setting 1) or in parallel with each other (setting 2). Ignore the resistance of the battery and the connecting wires.

(a) Calculate the combined resistance of the two elements:

(i) in setting 1 [1]

(ii) in setting 2. [1]

(b) Calculate the current from the battery in setting 2. [1]

(c) Calculate the power of the heater in setting 1 and in setting 2. [2]

(d) Determine the ratio power in setting 2 : power in setting 1. [1]

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22 · Part 3 Extended Action and use of circuit components [6]

A 12 V supply is connected across the two ends, X and Y, of a uniform resistance track. The total resistance of the track is 600 Ω. A sliding contact S can be moved along the track from X to Y. This arrangement is a variable potential divider. The output p.d. is measured by a voltmeter connected between X and S. The voltmeter has a very high resistance, so it takes no current.

(a) Describe how the output p.d. changes as S is moved slowly from X to Y. [2]

(b) S is positioned so that the resistance of the track between X and S is 150 Ω. Calculate the output p.d. [3]

(c) Suggest one use for a variable potential divider. [1]

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23 · Part 3 Extended Action and use of circuit components [6]

A light-dependent resistor (LDR) is connected in series with a 3.0 kΩ fixed resistor across a 6.0 V supply. The table shows the resistance of the LDR in two different lighting conditions.

lighting condition	resistance of LDR / kΩ
dark room	27
bright sunlight	1.0

(a) Calculate the p.d. across the LDR in the dark room. [2]

(b) Calculate the p.d. across the LDR in bright sunlight. [2]

(c) In bright sunlight, state which component has the larger p.d. across it. Explain your answer. [2]

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24 · Part 3 Core Electrical safety [5]

A washing machine has a metal outer casing. It is connected to the mains supply by a plug containing a fuse and a cable with live, neutral and earth wires. The earth wire is connected to the metal casing.

(a) A fault occurs and the live wire touches the metal casing. Explain how the earth wire and the fuse together protect a person who touches the casing. [3]

(b) A hairdryer has a plastic casing and is double-insulated. Its cable has no earth wire. Explain why the hairdryer is safe to use without an earth wire. [1]

(c) The plug of the hairdryer still contains a fuse. State what this fuse protects. [1]

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25 · Part 3 Core Electrical safety [7]

A kitchen circuit is connected to the 230 V mains supply. It is protected by a trip switch (circuit breaker) set at 20 A. The table shows the power of three appliances used on this circuit.

appliance	power / W
kettle	2760
toaster	920
microwave oven	1150

(a) Calculate the current in the kettle when it is in use. [2]

(b) All three appliances are used at the same time. Calculate the total current in the kitchen circuit. [2]

(c) Explain what happens to the trip switch when all three appliances are used at the same time, and why this makes the circuit safer. [2]

(d) State one advantage of a trip switch compared with a fuse. [1]

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26 · Part 3 Core Electrical safety [4]

A student plugs a four-way extension lead into a single wall socket. A fan heater, a kettle and a hairdryer are plugged into the extension lead and are all used at the same time. The cable of the hairdryer is old, and its insulation is damaged in one place.

(a) State the hazard caused by the damaged insulation. [1]

(b) Explain why using all three appliances at the same time through the extension lead is hazardous. [2]

(c) The mains cable to the kettle contains three wires. Name the three wires. [1]

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27 · Part 3 Core Electromagnetic induction [5]

A student is given a coil of insulated wire with many turns, a sensitive centre-zero ammeter, connecting leads and a bar magnet.

(a) Describe how the student can use this apparatus to demonstrate electromagnetic induction. Include what the student observes. [3]

(b) The magnet is then held at rest inside the coil. Explain why the ammeter reading is zero. [1]

(c) Describe another way to induce an e.m.f. in the coil using this apparatus while the magnet is kept still. [1]

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28 · Part 3 Core Electromagnetic induction [4]

A student investigates the e.m.f. induced in a coil when a bar magnet moves through it. She uses coils with different numbers of turns, and moves the same magnet through each coil at a low speed or at a high speed. The table shows her results.

test	number of turns	speed of magnet	maximum e.m.f. / V
1	300	low	0.40
2	600	low	0.80
3	300	high	1.10

(a) Using tests 1 and 2, state how the number of turns on the coil affects the induced e.m.f. [1]

(b) Using tests 1 and 3, state how the speed of the magnet affects the induced e.m.f. [1]

(c) Predict the maximum e.m.f. for a coil of 600 turns with the magnet moving at the same high speed as in test 3. [1]

(d) State one other factor, not tested in this investigation, that affects the magnitude of the induced e.m.f. [1]

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29 · Part 3 Extended Electromagnetic induction [5]

The north pole of a bar magnet is pushed towards end P of a coil. The coil is connected to a resistor so that there is a complete circuit, and an induced current flows in the coil.

(a) State the magnetic pole produced at end P of the coil by the induced current. Explain your answer. [2]

(b) The magnet is then pulled away from end P. State the magnetic pole now produced at end P. [1]

(c) Explain, in terms of energy, why the person must do work to push the magnet towards the coil. [2]

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30 · Part 3 Extended Electromagnetic induction [6]

A straight copper wire lies in the plane of the page, running from the bottom of the page to the top. A uniform magnetic field is directed into the page. The ends of the wire are connected by long flexible leads to a sensitive centre-zero ammeter outside the field.

(a) The wire is moved towards the left of the page, staying parallel to its original direction. State the direction of the induced current in the wire. [1]

(b) The wire is now moved towards the right of the page at the same speed. State how the ammeter reading compares with the reading in (a). [1]

(c) The wire is now moved towards the top of the page, along its own length. State and explain whether an e.m.f. is induced in the wire. [2]

(d) State two changes that would increase the magnitude of the induced e.m.f. when the wire is moved towards the left. [2]

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1 · Part 4 Extended The a.c. generator [1]

In a simple a.c. generator, a coil rotates between the poles of a magnet. The ends of the coil are connected to slip rings, and brushes press against the slip rings.

What is the purpose of the slip rings and brushes?

A to reverse the connections to the coil every half turn so that the output is d.c.

B to keep a continuous electrical connection between the rotating coil and the external circuit without the wires twisting

C to increase the strength of the magnetic field through the coil

D to make the coil rotate at a constant speed

2 · Part 4 Core Magnetic effect of a current [1]

There is a current in a long straight wire. The wire passes vertically through a hole in a horizontal sheet of card.

Which statement describes the shape of the magnetic field lines on the card?

A straight lines spreading outwards from the wire in all directions

B straight parallel lines crossing the card from one edge to the other

C a single loop running around the edge of the card

D circles centred on the wire

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3 · Part 4 Core Magnetic effect of a current [1]

There is a current in a solenoid that lies horizontally, with a left end and a right end. A student looks at the solenoid end-on from the left. From where the student is looking, the current flows anticlockwise around the coil.

Which statement about the magnetic poles of the solenoid is correct?

A The left end is a north pole and the right end is a south pole.

B The left end is a south pole and the right end is a north pole.

C Both ends are north poles.

D There are no poles because the magnetic field is only inside the solenoid.

4 · Part 4 Extended Magnetic effect of a current [1]

There is a constant current in a long straight wire. Point P is 2 cm from the wire. Point Q is 6 cm from the wire.

Which statement is correct?

A The magnetic field has the same strength at P and at Q, because the current is the same.

B The magnetic field is stronger at Q, because the circle through Q is larger.

C The magnetic field is stronger at P, because the field is stronger closer to the wire.

D There is no magnetic field at P or at Q, because the field is only inside the wire.

5 · Part 4 Core Magnetic effect of a current [1]

Using a relay, a small current in one circuit can switch on a large current in a second circuit.

Which statement describes how a relay works?

A A current in the coil magnetises a soft-iron core, which attracts an iron armature and closes the contacts in the second circuit.

B A current in the coil heats the contacts, which expand until they touch and close the second circuit.

C The current in the coil flows directly through the contacts and adds to the current in the second circuit.

D A current in the coil increases the voltage of the second circuit in the same way as a transformer.

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6 · Part 4 Core Force on a current-carrying conductor [1]

Between the poles of a magnet there is a horizontal copper wire. When there is a current in the wire, the wire is pushed upwards.

The current is reversed. At the same time, the magnet is turned round so that the direction of the magnetic field is also reversed.

What happens to the wire?

A There is no force on the wire.

B The wire is pushed upwards.

C The wire is pushed downwards.

D The wire is pushed along its own length.

7 · Part 4

Extended | Force on a current-carrying conductor | [1]

In this question, a straight wire lies across a page. The current in the wire flows from the left of the page to the right of the page. A uniform magnetic field acts into the page.

What is the direction of the force on the wire?

A. into the page

B. towards the bottom of the page

C. towards the right of the page

D. towards the top of the page

8 · Part 4

Extended | Force on a current-carrying conductor | [1]

In a vacuum, a narrow beam of electrons travels from the top of a page towards the bottom of the page. The beam passes through a uniform magnetic field that acts into the page.

In which direction is the beam deflected as it enters the field?

A. towards the right of the page

B. into the page

C. towards the left of the page

D. It is not deflected, because the electrons are not in a wire.

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9 · Part 4

Core | The d.c. motor | [1]

When a coil carrying a current is placed in a magnetic field, it experiences a turning effect.

Which change does **not** increase the turning effect on the coil?

A. increasing the number of turns on the coil

B. reversing the direction of the current

C. increasing the current

D. using a stronger magnet

10 · Part 4 Core The transformer [1]

Which description is correct for a simple transformer used to change the size of an alternating voltage?

A two coils of insulated wire wound on a soft-iron core

B two coils of bare copper wire wound on a steel core

C one coil of insulated wire wound around a permanent magnet

D two coils of insulated wire wound on a plastic tube

11 · Part 4 Core The transformer [1]

The transformer in this question has 1200 turns on its primary coil and 60 turns on its secondary coil. The primary coil is connected to a 120 V a.c. supply.

What is the voltage across the secondary coil?

A 2400 V

B 120 V

C 20 V

D 6.0 V

12 · Part 4 Core The transformer [1]

At a power station, a step-up transformer changes an alternating voltage of 25 kV to 400 kV. The primary coil has 800 turns.

How many turns are on the secondary coil?

A 50

B 16

C 12 800

D 320 000

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13 · Part 4 Core The transformer [1]

Electrical energy is transmitted across a country along cables at a very high voltage.

What is an advantage of using a high voltage?

A The current in the cables is smaller, so less energy is wasted as heat in the cables.

B The electrical energy travels along the cables at a higher speed.

C The current in the cables is larger, so more power reaches the consumers.

D High-voltage cables are safer for people who live near them.

14 · Part 4 Extended The transformer [1]

The transformer in this question is 100% efficient. Its primary coil is connected to a 230 V a.c. supply. The voltage across the secondary coil is 11.5 V and the current in the secondary coil is 4.0 A.

What is the current in the primary coil?

A 80 A

B 0.050 A

C 0.20 A

D 4.0 A

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15 · Part 4 Extended The transformer [1]

Electrical power is transmitted along a pair of cables. The transmission voltage is increased from 50 kV to 250 kV, but the power transmitted stays the same. The resistance of the cables does not change.

What happens to the power wasted as heat in the cables?

A It decreases by a factor of 5.

B It decreases by a factor of 25.

C It increases by a factor of 5.

D It increases by a factor of 25.

16 · Part 4 Extended The a.c. generator [5]

A student builds a simple a.c. generator. A rectangular coil of insulated wire is turned by a handle between the north pole and the south pole of two permanent magnets. Each end of the coil is joined to its own metal ring. Both rings are fixed to the axle and turn with the coil. A carbon block presses against each ring, and the two carbon blocks are connected by wires to a lamp.

(a) State the name of:

(i) the metal rings [1]

(ii) the carbon blocks. [1]

(b) Explain why the coil is connected to the lamp through the rings and carbon blocks, and not by wires joined directly to the ends of the coil. [2]

(c) A bicycle generator is a different design of a.c. generator. In this design a magnet rotates inside a coil that does not move.

Suggest why this design does not need rings and carbon blocks. [1]

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17 · Part 4

Core

Magnetic effect of a current

[6]

A student investigates the magnetic field around a long straight wire. The wire is vertical and passes through a small hole in the centre of a horizontal sheet of card. The wire is connected in series with a switch and a d.c. power supply that can provide a large current.

(a) Describe how the student can use iron filings to show the pattern of the magnetic field on the card. [2]

(b) Describe the pattern that the student sees. [1]

(c) Describe how the student can find the direction of the magnetic field at different places on the card. [2]

(d) The current in the wire flows downwards, through the card from above. The student looks down on the card from above.

State whether the magnetic field lines around the wire are clockwise or anticlockwise, as seen by the student. [1]

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18 · Part 4 Extended Magnetic effect of a current [6]

A solenoid is made by winding insulated wire around a long cardboard tube. One end of the solenoid is labelled X. The solenoid is connected to a d.c. power supply.

A plotting compass is placed on the axis of the solenoid, just outside end X. The north-seeking end of the compass needle points away from the solenoid.

(a) State whether end X is a north pole or a south pole. [1]

(b)

(i) Describe the magnetic field lines inside the solenoid, away from its ends. [1]

(ii) State how the strength of the magnetic field inside the solenoid compares with the strength of the field just outside the side of the solenoid. [1]

(c) The current in the solenoid is increased. State the effect on the magnetic field. [1]

(d) The connections to the power supply are now reversed, so that the current flows in the opposite direction. Describe the effect of this on the magnetic field of the solenoid and on end X. [2]

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19 · Part 4

Core

Magnetic effect of a current

[5]

A moving-coil loudspeaker contains a light coil of wire fixed to a paper cone. The coil fits around one pole of a strong permanent magnet, so the coil is in a magnetic field. An amplifier passes an alternating current through the coil.

(a) Explain why there is a force on the coil when there is a current in it. [1]

(b) Explain how the alternating current in the coil causes the loudspeaker to produce a sound. [3]

(c) The frequency of the alternating current is increased. State what happens to the sound produced. [1]

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20 · Part 4 Core Magnetic effect of a current [5]

The starter motor of a car needs a current of 150 A. The driver starts the car by turning a key switch. The key switch is in a separate circuit that carries a current of only 0.20 A. This small-current circuit contains the coil of a relay. The contacts of the relay are in the starter motor circuit.

(a) Suggest why the starter motor is not switched on and off directly by the key switch. [1]

(b) Describe how turning the key switch makes the starter motor work. [3]

(c) Give one other application of a relay. [1]

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21 · Part 4 Core Force on a current-carrying conductor [6]

A student sets up an experiment to show that a force acts on a current-carrying conductor in a magnetic field. Two straight, parallel, horizontal copper rails are connected to a d.c. power supply and a switch. A short, light copper rod rests across the two rails, so it completes the circuit. A strong U-shaped magnet is placed so that the rod lies between its poles and the magnetic field is vertical.

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(a) The switch is closed. Describe what the student observes. [1]

(b) Explain why this happens. [2]

(c) The student reverses the connections to the power supply and closes the switch again. State what the student observes. [1]

(d) With the connections still reversed, the student also turns the magnet over so that the magnetic field is reversed. State how the movement of the rod now compares with the movement in (a). [1]

(e) The student replaces the copper rod with a glass rod of the same size. State and explain what happens when the switch is closed. [1]

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22 · Part 4 Extended Force on a current-carrying conductor [6]

In this question, give each direction as one of: towards the top of the page, towards the bottom of the page, towards the left of the page, towards the right of the page, into the page, out of the page.

(a) A straight wire lies in the plane of the page. The current in the wire flows from the top of the page to the bottom of the page. A uniform magnetic field acts from the left of the page to the right of the page.

State the direction of the force on the wire. [1]

(b) The current in the wire is unchanged. The magnetic field is changed so that it now acts into the page.

State the direction of the force on the wire. [1]

(c) A narrow beam of α-particles travels in a vacuum from the left of the page to the right of the page. It enters a uniform magnetic field that acts out of the page.

(i) State the direction of the force on the α-particles as they enter the field. [1]

(ii) A beam of β-particles travels in the same direction through the same field. State the direction of the force on the β-particles and explain why it is different from your answer to (c)(i). [2]

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(iii) A beam of γ-rays travels in the same direction through the same field. State and explain what happens to the beam. [1]

23 · Part 4 Core The d.c. motor [4]

A rectangular coil of wire is placed between the poles of a magnet. When there is a current in the coil, the coil experiences a turning effect.

(a) State three changes that would each increase the turning effect on the coil. [3]

(b) The direction of the current in the coil is reversed. State the effect on the direction in which the coil turns. [1]

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24 · Part 4 Extended The d.c. motor [6]

A simple d.c. motor has a rectangular coil mounted on an axle between the north pole and the south pole of a magnet. The magnetic field is horizontal. The axle is horizontal and at right angles to the field. The current enters and leaves the coil through two carbon brushes that press against a split-ring commutator fixed to the axle.

(a) The coil starts with its plane horizontal, parallel to the magnetic field. The current is switched on.

Explain why the coil starts to rotate. [2]

(b) Explain why the split-ring commutator is needed for the coil to keep rotating in the same direction. [3]

(c) State the purpose of the brushes. [1]

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25 · Part 4 Core The transformer [5]

(a) Describe the construction of a simple transformer. [3]

(b) A transformer has 500 turns on its primary coil and 2000 turns on its secondary coil.

State whether this is a step-up transformer or a step-down transformer. Give a reason for your answer. [2]

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26 · Part 4 Core The transformer [6]

A laboratory power pack contains a transformer. The primary coil has 1840 turns and is connected to the 230 V a.c. mains supply.

(a) Calculate the number of turns needed on a secondary coil to give an output voltage of 12 V. [3]

(b) The power pack has a second secondary coil with 24 turns. Calculate the output voltage of this coil. [2]

(c) State whether the transformer in the power pack is a step-up transformer or a step-down transformer. [1]

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27 · Part 4 Core The transformer [5]

A power station generates electricity at an alternating voltage of 20 kV. The electricity is transmitted across the country along overhead cables at 275 kV. It is supplied to homes at 230 V.

(a) State the type of transformer used:

(i) at the power station [1]

(ii) near the homes. [1]

(b) State two advantages of transmitting the electricity at a high voltage. [2]

(c) Suggest why the voltage is reduced before the electricity is supplied to homes. [1]

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28 · Part 4 Extended The transformer [5]

A student connects the primary coil of an iron-cored transformer to a low-voltage a.c. supply. A lamp is connected across the secondary coil. There is no electrical connection between the two coils, but the lamp lights.

(a) Explain how a voltage is produced across the secondary coil. [3]

(b) The student replaces the a.c. supply with a battery and a switch. When the switch is closed, the lamp flashes very briefly and then stays off, even though there is a steady current in the primary coil.

Explain why the lamp does not stay lit. [2]

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29 · Part 4 Extended The transformer [6]

A transformer is used to operate a lamp rated 24 V, 60 W from a 240 V a.c. supply. The lamp is connected across the secondary coil and works at its normal brightness. The transformer is 100% efficient.

(a) Calculate the current in the lamp. [2]

(b) Calculate the current in the primary coil. [3]

(c) State the ratio of the number of turns on the primary coil to the number of turns on the secondary coil. [1]

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30 · Part 4 Extended The transformer [8]

A small wind farm transmits 600 kW of electrical power to a town along cables with a total resistance of 5.0 Ω. The engineers compare transmitting this power at 10 kV and at 30 kV.

(a) Calculate the current in the cables when the power is transmitted at 10 kV. [2]

(b) Calculate the power wasted as heat in the cables when the power is transmitted at 10 kV. [2]

(c) Calculate the power wasted as heat in the cables when the power is transmitted at 30 kV. [2]

(d) Use your answers to explain why electrical power is transmitted at high voltages. [2]

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